



Scalable Gibbs sampler using randomized sketching for massive and high-dimensional Bayesian regression

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Introduction

- High-dimensional Bayesian regression with spike-and-slab priors offers a unified framework for variable selection, estimation, and model uncertainty quantification via Gibbs sampling.
- However, standard Gibbs sampling becomes **computationally prohibitive when the number of predictors p is large** since its computational complexity is $O(p^3)$ per iteration.
- Many existing scalable Bayesian methods for high-dimensional regression are tailored to the $n \ll p$ setting and may still face **computational bottlenecks when $n \gg p$** .
- To address these computational challenges, we propose a novel sketching-based approach for **scalable Bayesian inference** in massive, high-dimensional settings where $n \gg p$ but p remains large.

Bayesian regression with spike-and-slab priors

- The Bayesian regression model with spike-and-slab priors can be expressed as the following hierarchical representation (George and McCulloch 1993):

$$\begin{aligned} \text{[Likelihood]} \quad & \mathbf{y} \mid \boldsymbol{\beta}, \sigma^2 \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n), \\ \text{[Prior]} \quad & \beta_j \mid \sigma^2, \gamma_j \stackrel{\text{ind}}{\sim} N(0, \sigma^2 d_{\gamma_j}), \\ & \sigma^2 \sim \text{IG}(a_0, b_0), \\ & \gamma_j \stackrel{\text{iid}}{\sim} \text{Ber}(w), \end{aligned}$$

where w is the prior inclusion probability, $d_{\gamma_j} = (1 - \gamma_j)\tau_0^2 + \gamma_j\tau_1^2$ with $d_0 = \tau_0^2$ and $d_1 = \tau_1^2$, and $\mathbf{D}_\gamma = \text{diag}(d_{\gamma_1}, \dots, d_{\gamma_p})$. Let $\boldsymbol{\Sigma}_\gamma^{-1} = \mathbf{X}^\top \mathbf{X} + \mathbf{D}_\gamma^{-1}$ denote the unscaled conditional precision matrix; then $\boldsymbol{\Sigma}_\gamma$ is the unscaled conditional covariance matrix.

- Posterior sampling can be performed by iterating the following Gibbs sampling steps:

Step 1: Generate $\boldsymbol{\beta}$ from

$$\boldsymbol{\beta} \mid \mathbf{y}, \sigma^2, \boldsymbol{\gamma} \sim N(\boldsymbol{\Sigma}_\gamma \mathbf{X}^\top \mathbf{y}, \sigma^2 \boldsymbol{\Sigma}_\gamma).$$

Step 2: Generate σ^2 from

$$\sigma^2 \mid \mathbf{y}, \boldsymbol{\beta}, \boldsymbol{\gamma} \sim \text{IG}\left(a_0 + \frac{n+p}{2}, b_0 + \frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \boldsymbol{\beta}^\top \mathbf{D}_\gamma^{-1} \boldsymbol{\beta}}{2}\right).$$

Step 3: Generate γ_j from

$$\gamma_j \mid \mathbf{y}, \sigma^2, \boldsymbol{\beta} \stackrel{\text{ind}}{\sim} \text{Ber}(p_j), \quad p_j = \frac{w\phi(\beta_j; 0, \sigma^2 d_1)}{w\phi(\beta_j; 0, \sigma^2 d_1) + (1-w)\phi(\beta_j; 0, \sigma^2 d_0)},$$

where $\phi(\cdot; 0, v)$ denotes the $N(0, v)$ density.

- * Remark:** Computing $\boldsymbol{\Sigma}_\gamma = (\mathbf{X}^\top \mathbf{X} + \mathbf{D}_\gamma^{-1})^{-1}$ incurs a computational cost of $O(p^3)$ per iteration, which becomes prohibitively expensive as p increases.

Existing scalable Gibbs samplers

- Efficient Gaussian sampler.** (Bhattacharya, Chakraborty, and Mallick 2016) The Gaussian update can avoid direct $p \times p$ Cholesky decomposition by using the **Woodbury identity**:

$$(\mathbf{X}^\top \mathbf{X} + \mathbf{D}_\gamma^{-1})^{-1} = \mathbf{D}_\gamma - \mathbf{D}_\gamma \mathbf{X}^\top (\mathbf{I}_n + \mathbf{X} \mathbf{D}_\gamma \mathbf{X}^\top)^{-1} \mathbf{X} \mathbf{D}_\gamma.$$
This reduces the main Gaussian update to an $n \times n$ linear system, with cost about $O(n^2 p)$.
- Scalable Spike-and-Slab, S^3 .** (Biswas, Mackey, and Meng 2022) The S^3 sampler updates the active-set covariance incrementally as predictors enter or leave. **Schur-complement** add/delete updates avoid repeated full matrix construction, giving about $O(np)$ cost per iteration for linear regression.
- Remaining bottleneck.** These methods are mainly useful in the $n \ll p$ setting. When $n \approx p$ or $n > p$ with large p , they can still be inefficient because each iteration uses full-data terms.
- Sketched data approach.** (Guhaniyogi and Scheffler 2025) These samplers can be applied by replacing $(\mathbf{X}, \mathbf{y})$ with sketched data $(\tilde{\mathbf{X}}, \tilde{\mathbf{y}}) = (\mathbf{S}\mathbf{X}, \mathbf{S}\mathbf{y})$, where $\mathbf{S} \in \mathbb{R}^{k \times n}$ with $S_{ij} \sim N(0, 1/k)$. This reduces n to k , but approximating the exact Gibbs sampler at the density level can be difficult.

Proposed method: block-wise sketched Gibbs sampler

- Given the current indicator vector $\boldsymbol{\gamma}$, define the active and inactive sets

$$\mathcal{A} = \{j : \gamma_j = 1\}, \quad \mathcal{I} = \{j : \gamma_j = 0\}.$$

Accordingly, partition the design matrix and regression coefficients as

$$\mathbf{X} = [\mathbf{X}_\mathcal{A} \ \mathbf{X}_\mathcal{I}], \quad \boldsymbol{\beta} = (\boldsymbol{\beta}_\mathcal{A}^\top, \boldsymbol{\beta}_\mathcal{I}^\top)^\top.$$

- To reduce the size of the data, we use a Gaussian sketching matrix $\mathbf{S} \in \mathbb{R}^{k \times n}$ with $S_{ij} \sim N(0, 1/k)$ and form the sketched data $\tilde{\mathbf{X}} = \mathbf{S}\mathbf{X}$ and $\tilde{\mathbf{y}} = \mathbf{S}\mathbf{y}$, where $k \ll \min(n, p)$.
- Instead of sketching all conditional updates, the proposed sampler **applies sketching selectively** to the computationally expensive **inactive covariance update** and the **global variance update**.

Algorithm: Overall block-wise sketched Gibbs sampler

1. Update the regression coefficients:

$$\boldsymbol{\beta}_\mathcal{I} \mid \mathbf{y}, \sigma^2, \boldsymbol{\beta}_\mathcal{A}, \boldsymbol{\gamma} \sim N(\underbrace{\boldsymbol{\Sigma}_\mathcal{I} \mathbf{X}_\mathcal{I}^\top (\mathbf{y} - \mathbf{X}_\mathcal{A} \boldsymbol{\beta}_\mathcal{A})}_{\text{via Algorithm 1}}, \underbrace{\sigma^2 \tilde{\boldsymbol{\Sigma}}_\mathcal{I}}_{\text{via Algorithm 2}}), \quad \tilde{\boldsymbol{\Sigma}}_\mathcal{I}^{-1} = \tilde{\mathbf{X}}_\mathcal{I}^\top \tilde{\mathbf{X}}_\mathcal{I} + \tau_0^{-2} \mathbf{I},$$

$$\boldsymbol{\beta}_\mathcal{A} \mid \mathbf{y}, \sigma^2, \boldsymbol{\beta}_\mathcal{I}, \boldsymbol{\gamma} \sim N(\underbrace{\boldsymbol{\Sigma}_\mathcal{A} \mathbf{X}_\mathcal{A}^\top (\mathbf{y} - \mathbf{X}_\mathcal{I} \boldsymbol{\beta}_\mathcal{I})}_{\text{via Algorithm 3}}, \sigma^2 \boldsymbol{\Sigma}_\mathcal{A}).$$

2. Update the variance:

$$\sigma^2 \mid \mathbf{y}, \boldsymbol{\beta}, \boldsymbol{\gamma} \sim \text{IG}\left(a_0 + \frac{n+p}{2}, b_0 + \frac{\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|_2^2 + \boldsymbol{\beta}^\top \mathbf{D}_\gamma^{-1} \boldsymbol{\beta}}{2}\right).$$

3. Update the latent indicators:

$$\gamma_j \mid \mathbf{y}, \sigma^2, \boldsymbol{\beta} \stackrel{\text{ind}}{\sim} \text{Ber}(p_j).$$

Algorithms for block-wise updates

$\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$: **update** $([\mathbf{G}_\ell]_{\mathcal{A}, \mathcal{A}})^{-1}$ via **Schur complement**, where δ_ℓ is the number of changed indicators.

Algorithm 1: Exact inactive mean update

$\mathcal{O}(p|\mathcal{A}|)$

Input: current active set $\mathcal{A}$ and active coefficients $\boldsymbol{\beta}_\mathcal{A}$.

Precompute: $\mathbf{H} = \mathbf{X}\mathbf{X}^\top + \tau_0^{-2} \mathbf{I}_n$, $\mathbf{M} = \mathbf{X}^\top \mathbf{H}^{-1} \mathbf{X}$, $\mathbf{v} = \mathbf{X}^\top \mathbf{H}^{-1} \mathbf{y}$, $\mathbf{G}_1 = \mathbf{I}_p - \mathbf{M}$.

Output: $\boldsymbol{\mu}_\mathcal{I} = (\mathbf{v}_\mathcal{I} - \mathbf{M}_{\mathcal{I}, \mathcal{A}} \boldsymbol{\beta}_\mathcal{A}) + \mathbf{M}_{\mathcal{I}, \mathcal{A}} ([\mathbf{G}_1]_{\mathcal{A}, \mathcal{A}})^{-1} (\mathbf{v}_\mathcal{A} - \mathbf{M}_{\mathcal{A}, \mathcal{A}} \boldsymbol{\beta}_\mathcal{A})$.

Algorithm 2: Sketched inactive covariance sampling

$\mathcal{O}(kp)$

Input: exact inactive mean $\boldsymbol{\mu}_\mathcal{I}$ and current active set $\mathcal{A}$.

Precompute: $\tilde{\mathbf{X}} = \mathbf{S}\mathbf{X}$, $\mathbf{M}_0 = \tau_0^{-2} \mathbf{I}_k + \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top$, $\mathbf{G}_2 = \mathbf{I}_p - \tilde{\mathbf{X}}^\top \mathbf{M}_0^{-1} \tilde{\mathbf{X}}$.

Noise: sample $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_k)$, $\boldsymbol{\xi} \sim N(\mathbf{0}, \mathbf{I}_{|\mathcal{I}|})$, and set $\mathbf{z}_\mathcal{I} = \tilde{\mathbf{X}}_\mathcal{I}^\top \mathbf{u} + \tau_0^{-1} \boldsymbol{\xi}$.

Compute: $\tilde{\eta}_\mathcal{I} = \tau_0^2 \mathbf{z}_\mathcal{I} - \tau_0^2 \tilde{\mathbf{X}}_\mathcal{I}^\top \mathbf{M}_0^{-1} (\tilde{\mathbf{X}}_\mathcal{I} \mathbf{z}_\mathcal{I}) - \tau_0^2 \tilde{\mathbf{X}}_\mathcal{I}^\top \mathbf{M}_0^{-1} \tilde{\mathbf{X}}_\mathcal{A} ([\mathbf{G}_2]_{\mathcal{A}, \mathcal{A}})^{-1} \tilde{\mathbf{X}}_\mathcal{A}^\top \mathbf{M}_0^{-1} (\tilde{\mathbf{X}}_\mathcal{I} \mathbf{z}_\mathcal{I})$.

Output: $\boldsymbol{\beta}_\mathcal{I} = \boldsymbol{\mu}_\mathcal{I} + \sigma \tilde{\eta}_\mathcal{I}$, $\text{Var}(\tilde{\eta}_\mathcal{I}) = \tilde{\boldsymbol{\Sigma}}_\mathcal{I}$.

Algorithm 3: Exact active block update

$\mathcal{O}(p|\mathcal{A}|)$

Input: newly sampled inactive coefficients $\boldsymbol{\beta}_\mathcal{I}$ and current active set $\mathcal{A}$.

Precompute: $\mathbf{G}_3 = \mathbf{X}^\top \mathbf{X} + \tau_1^{-2} \mathbf{I}_p$.

Noise: sample $\boldsymbol{\zeta}_\mathcal{A} \sim N(\mathbf{0}, [\mathbf{G}_3]_{\mathcal{A}, \mathcal{A}})$.

Output: $\boldsymbol{\beta}_\mathcal{A} = ([\mathbf{G}_3]_{\mathcal{A}, \mathcal{A}})^{-1} (\mathbf{r}_\mathcal{A}^* + \sigma \boldsymbol{\zeta}_\mathcal{A})$, $\mathbf{r}_\mathcal{A}^* = (\mathbf{X}^\top \mathbf{y})_\mathcal{A} - (\mathbf{X}^\top \mathbf{X})_{\mathcal{A}, \mathcal{I}} \boldsymbol{\beta}_\mathcal{I}$.

- Overall per-iteration cost:** $\mathcal{O}(kp + p|\mathcal{A}| + |\mathcal{A}|^2 \delta_\ell + \delta_\ell^3)$ using Schur-complement updates for δ_ℓ changed variables. Typically **dominated by $\mathcal{O}(kp)$** when $|\mathcal{A}| \ll k \ll p$.

Theorem 1. Inactive covariance approximation error bound

Let $\boldsymbol{\Sigma}_\mathcal{I}^{-1} = \mathbf{X}_\mathcal{I}^\top \mathbf{X}_\mathcal{I} + \tau_0^{-2} \mathbf{I}_{|\mathcal{I}|}$ and $\tilde{\boldsymbol{\Sigma}}_\mathcal{I}^{-1} = \tilde{\mathbf{X}}_\mathcal{I}^\top \tilde{\mathbf{X}}_\mathcal{I} + \tau_0^{-2} \mathbf{I}_{|\mathcal{I}|}$. Treating $\mathcal{I}$ as fixed independently of $\mathbf{S}$,

$$\mathbb{E}_\mathbf{S} \|\tilde{\boldsymbol{\Sigma}}_\mathcal{I} - \boldsymbol{\Sigma}_\mathcal{I}\|_F^2 \leq \frac{\tau_0^8}{k} \left[\{\text{tr}(\mathbf{X}_\mathcal{I}^\top \mathbf{X}_\mathcal{I})\}^2 + \|\mathbf{X}_\mathcal{I}^\top \mathbf{X}_\mathcal{I}\|_F^2 \right].$$

If $\lambda_{\max}(n^{-1} \mathbf{X}_\mathcal{I}^\top \mathbf{X}_\mathcal{I}) = O(1)$ and the shrinking-spike condition of Narisetty and He (2014), $n\tau_0^2 = o(1)$, holds, then

$$\mathbb{E}_\mathbf{S} \|\tilde{\boldsymbol{\Sigma}}_\mathcal{I} - \boldsymbol{\Sigma}_\mathcal{I}\|_F^2 = O\left(\frac{\tau_0^8 n^2 r_\mathcal{I}^2}{k}\right) = o(1), \quad r_\mathcal{I} = \text{rank}(\mathbf{X}_\mathcal{I}).$$

Theorem 2. Sketched residual quadratic form bound

For a fixed $\boldsymbol{\beta}$ and residual $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$ independent of $\mathbf{S}$, for any $0 < \epsilon < 1$,

$$\mathbb{P}\left(\left|\frac{\|\mathbf{S}\mathbf{r}\|_2^2}{\|\mathbf{r}\|_2^2} - 1\right| \leq \epsilon\right) \geq 1 - 2 \exp\left(-\frac{k\epsilon^2}{8}\right).$$

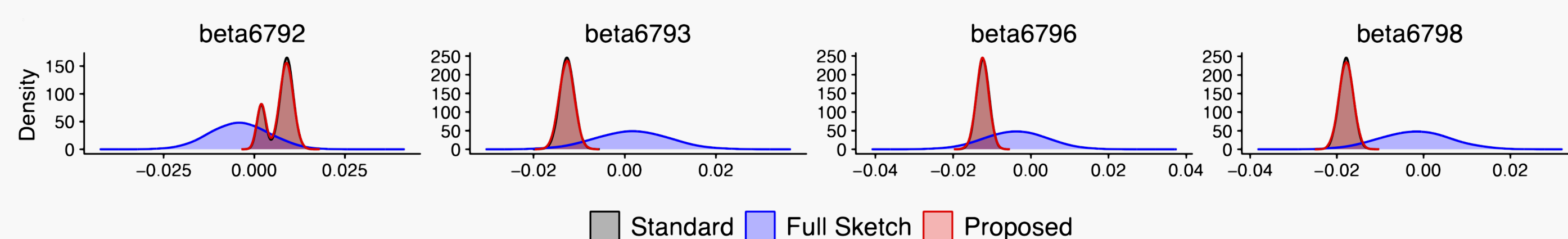
Simulation study

- Data: $n = 10,000$, $p \in \{1000, 2000\}$, AR(1) covariance $\Sigma_{ij} = 0.5^{|i-j|}$, and SNR = 1. True signals have $s_0 = 20$ nonzero coefficients with $\beta_j \sim U(0.1, 10)$; $\tau_0^2 = n^{-1.1}$, $\tau_1^2 = p^{2.2}/n$; 10 reps.
- Methods: exact Gibbs, full sketching (Guhaniyogi and Scheffler 2025) with S^3 (Biswas, Mackey, and Meng 2022), and proposed sampler. PIP denotes posterior inclusion probability.

p	k	Method	Time (s)	Speedup	PIP RMSE	Jaccard
1000	–	Exact	7678.9 (10.3) (2.1 h)	–	–	–
	100	Full sketch	2.3 (0.0)	3339×	0.127 (0.001)	0.06 (0.01)
	100	Proposed	4.9 (0.1)	1567×	0.009 (0.002)	0.96 (0.01)
	150	Full sketch	3.3 (0.2)	2327×	0.126 (0.001)	0.07 (0.01)
	150	Proposed	5.8 (0.2)	1324×	0.012 (0.003)	0.97 (0.01)
	–	Exact	60978.8 (284.4) (16.9 h)	–	–	–
2000	100	Full sketch	4.2 (0.2)	14519×	0.087 (0.001)	0.06 (0.00)
	100	Proposed	20.9 (0.9)	2918×	0.011 (0.004)	0.95 (0.02)
	200	Full sketch	8.4 (0.5)	7259×	0.086 (0.001)	0.09 (0.01)
	200	Proposed	23.8 (1.2)	2562×	0.010 (0.003)	0.95 (0.02)

Real data application: MovieLens 1M

- Movie ratings data: sampled $n = 290,580$ ratings and encoded $p = 7,911$ user/movie features.
- Used sketch size $k = 2,000$ with $\tau_0^2 = n^{-1.1}$ and $\tau_1^2 = p^{2.2}/n$.



Runtime. Exact Gibbs: 99,580.4 sec (≈ 27.7 h); proposed: 2,142.6 sec (≈ 35.7 min), 46.4× faster.

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